

Exposure to nitrosamines in thirdhand tobacco smoke increases cancer risk in non-smokers.

In addition to passive inhalation, non-smokers, and especially children, are exposed to residual tobacco smoke gases and particles that are deposited to surfaces and dust, known as thirdhand smoke (THS). However, until now the potential cancer risks of this pathway of exposure have been highly uncertain and not considered in public health policy. In this study, we estimate for the first time the potential cancer risk by age group through non-dietary ingestion and dermal exposure to carcinogen N-nitrosamines and tobacco-specific nitrosamines (TSNAs) measured in house dust samples. Using a highly sensitive and selective analytical approach we have determined the presence of nicotine, eight N-nitrosamines and five tobacco-specific nitrosamines in forty-six settled dust samples from homes occupied by both smokers and non-smokers. Using observations of house dust composition, we have estimated the cancer risk by applying the most recent official toxicological information. Calculated cancer risks through exposure to the observed levels of TSNAs at an early life stage (1 to 6years old) exceeded the upper-bound risk recommended by the USEPA in 77% of smokers' and 64% of non-smokers' homes. The maximum risk from exposure to all nitrosamines measured in a smoker occupied home was one excess cancer case per one thousand population exposed. The results presented here highlight the potentially severe long-term consequences of THS exposure, particularly to children, and give strong evidence of its potential health risk and, therefore, they should be considered when developing future environmental and health policies. Copyright © 2014 Elsevier Ltd. All rights reserved.